EDA, or **Exploratory Data Analysis**, is the process of analysing and visualising data to understand its structure, patterns, relationships, and anomalies.

It is done to know in and out of data, to decide what course of action to follow,

how our data is related, behaviour of data etc.

Categorical or numerical.

EDA:

- Univariate Analysis (numerical column / categorical column)
- Bivariate Analysis

  (numerical-numerical, categorical-categorical, numerical-categorical)
- Multivariate Analysis

UNIVARIATE ANALYSIS [Numerical column]

- Min, Max, Avg, Std.
- Null values
- Percentiles
- Outliers
- Histogram

UNIVARIATE ANALYSIS [Categorical column]

- Null values
- Pie chart or frequency count

BIVARIATE ANALYSIS [Numerical - Numerical]

- Scatter plot
- Covariance

$$\mathrm{Cov}(X,Y) = \frac{\sum(X_i-\bar{X})(Y_i-\bar{Y})}{n-1}$$

- Correlation

$$Correlation = \frac{Cov\,(x,y)}{\sigma x * \sigma y}$$

BIVARIATE ANALYSIS [Categorical - Categorical]

- Contingency Table

BIVARIATE ANALYSIS [Numerical - Categorical]

- Numerical analysis along categorical column